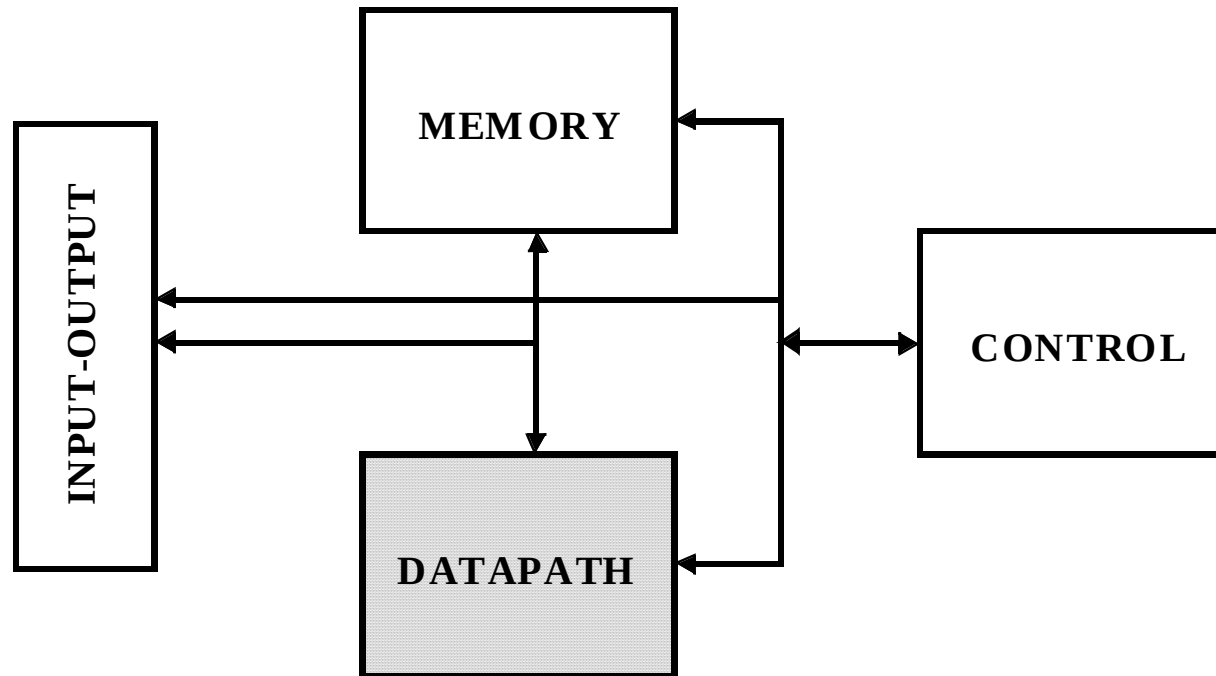


A Generic Digital Processor



Building Blocks for Digital Architectures

Arithmetic unit

- Bit-sliced datapath (adder, multiplier, shifter, comparator, etc.)

Memory

- RAM, ROM, Buffers, Shift registers

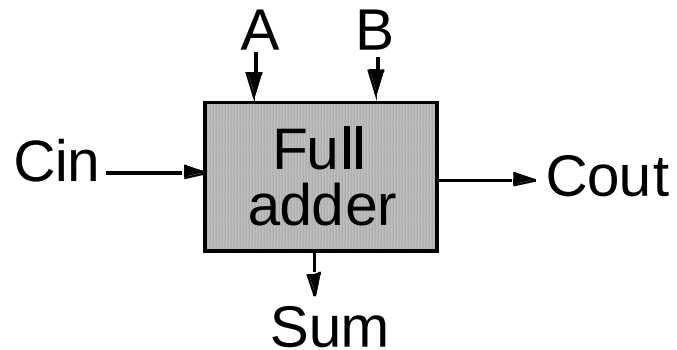
Control

- Finite state machine (PLA, random logic.)
- Counters

Interconnect

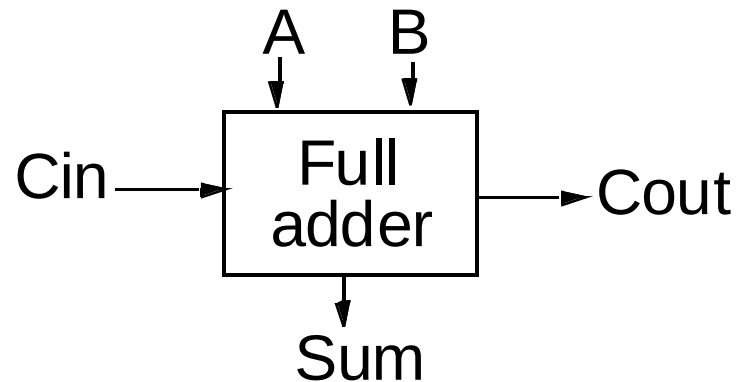
- Switches
- Arbiters
- Bus

Full-Adder



A	B	C_i	S	C_o	<i>Carry status</i>
0	0	0	0	0	delete
0	0	1	1	0	delete
0	1	0	1	0	propagate
0	1	1	0	1	propagate
1	0	0	1	0	propagate
1	0	1	0	1	propagate
1	1	0	0	1	generate
1	1	1	1	1	generate

The Binary Adder



$$\begin{aligned} S &= A \oplus B \oplus C_i \\ &= A\bar{B}\bar{C}_i + \bar{A}B\bar{C}_i + \bar{A}\bar{B}C_i + ABC_i \\ C_o &= AB + BC_i + AC_i \end{aligned}$$

Express Sum and Carry as a function of P, G, D

Define 3 new variable which ONLY depend on A, B

$$\text{Generate } (G) = AB$$

$$\text{Propagate } (P) = A \oplus B$$

$$\text{Delete} = \bar{A} \bar{B}$$

$$C_o(G, P) = G + PC_i$$

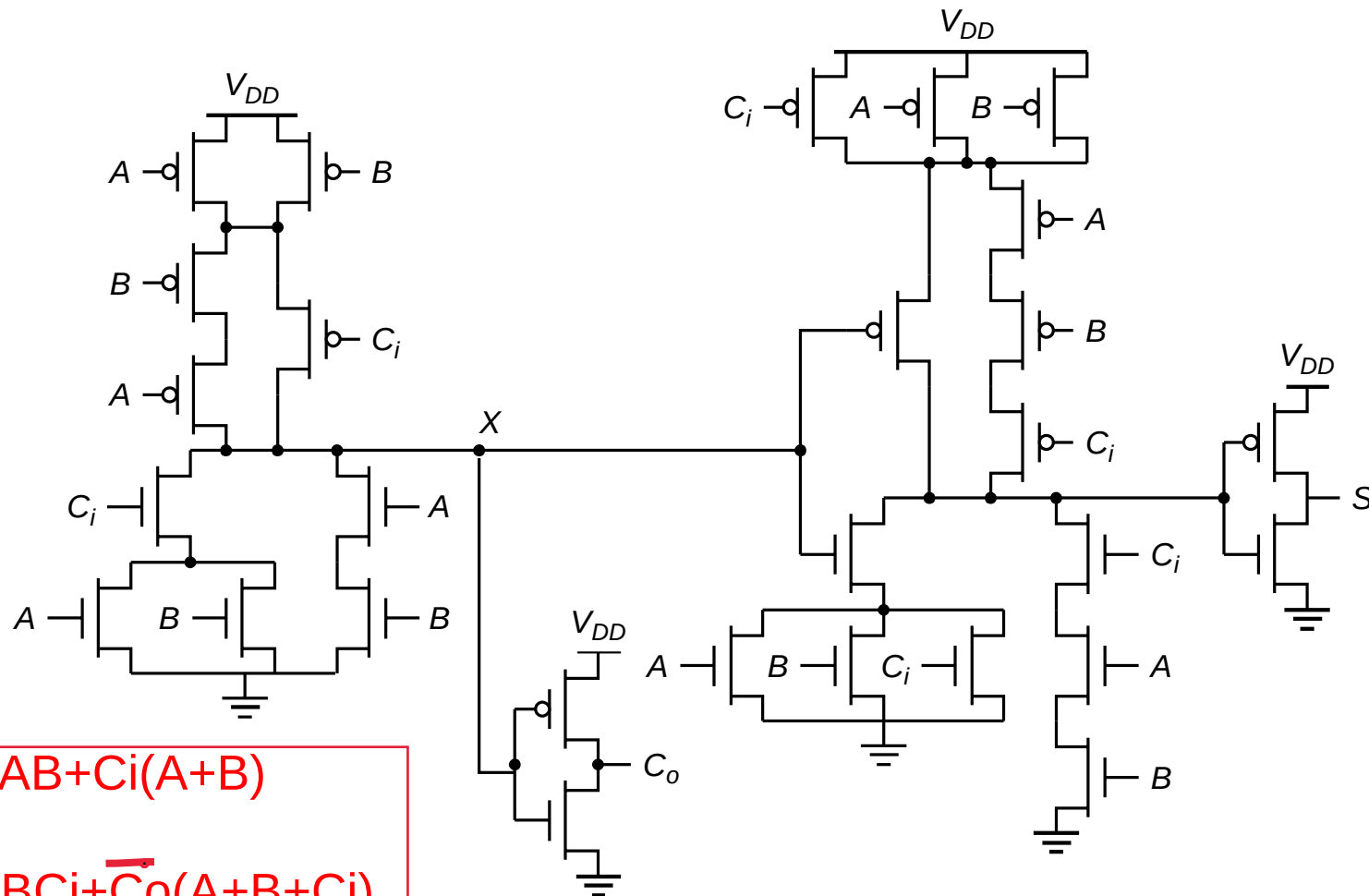
$$S(G, P) = P \oplus C_i$$

Can also derive expressions for S and C_o based on D and P

Note that we will be sometimes using an alternate definition for

$$\text{Propagate } (P) = A + B$$

Complimentary Static CMOS Full Adder



$$C_o = AB + C_i(A + B)$$

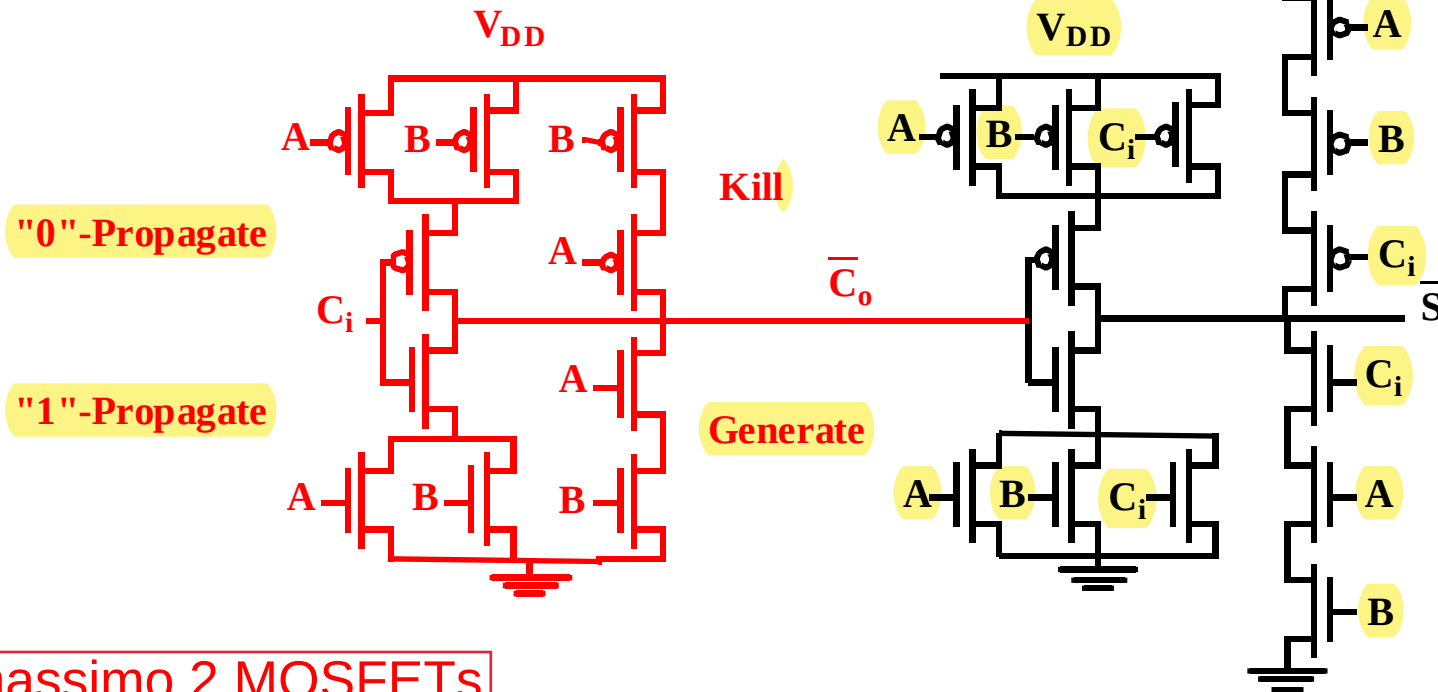
$$S = ABC_i + \overline{C_o}(A + B + C_i)$$

28 Transistors

A Better Structure: The Mirror Adder

$Co=1$ se $Generate=1$
 $Co=0$ se $Kill=1$
 $Co=C_i(A+B)$ altrimenti

$$S = ABC_i + \overline{Co}(A+B+C_i)$$



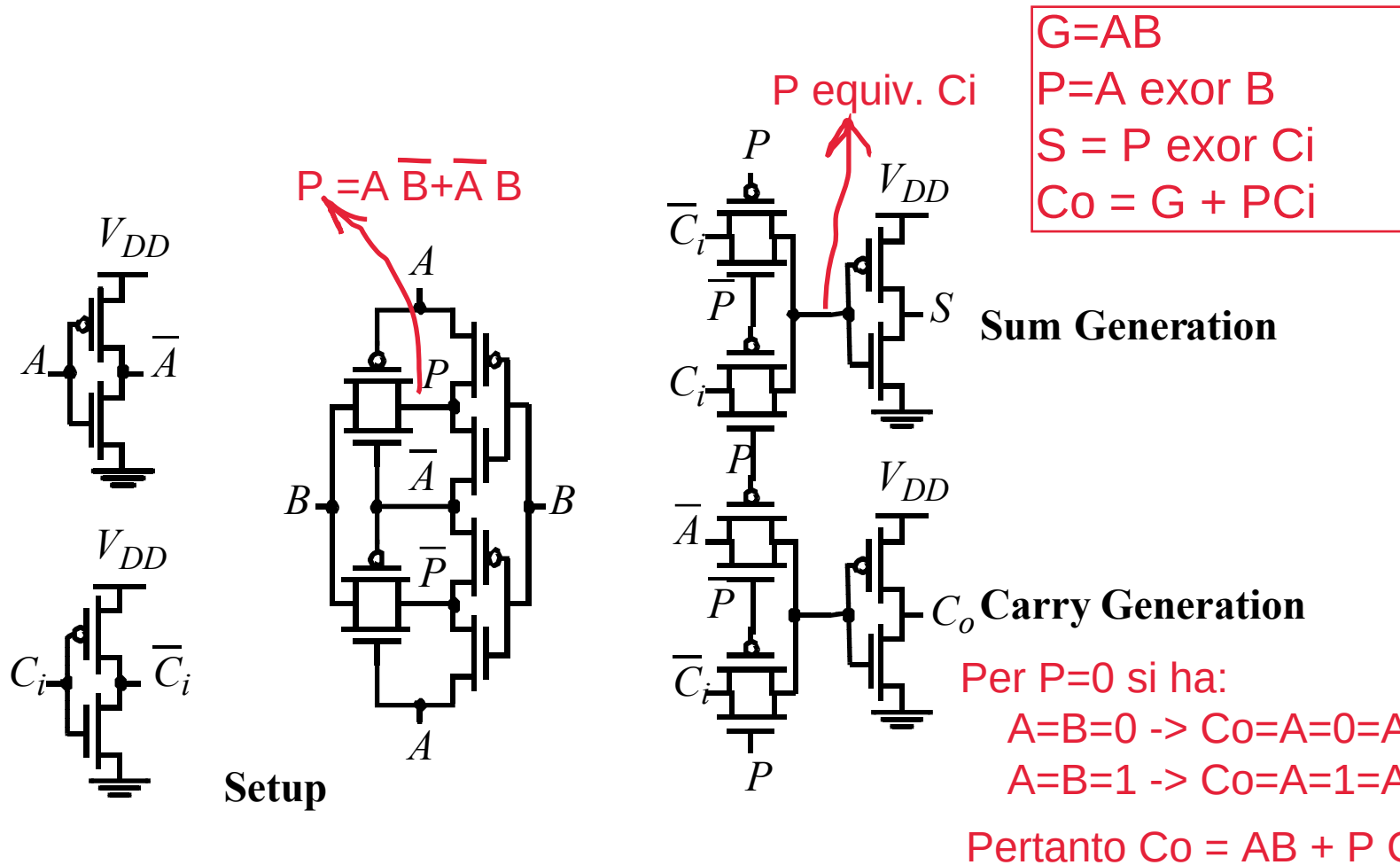
Al massimo 2 MOSFETs
in serie per il Co

24 transistors

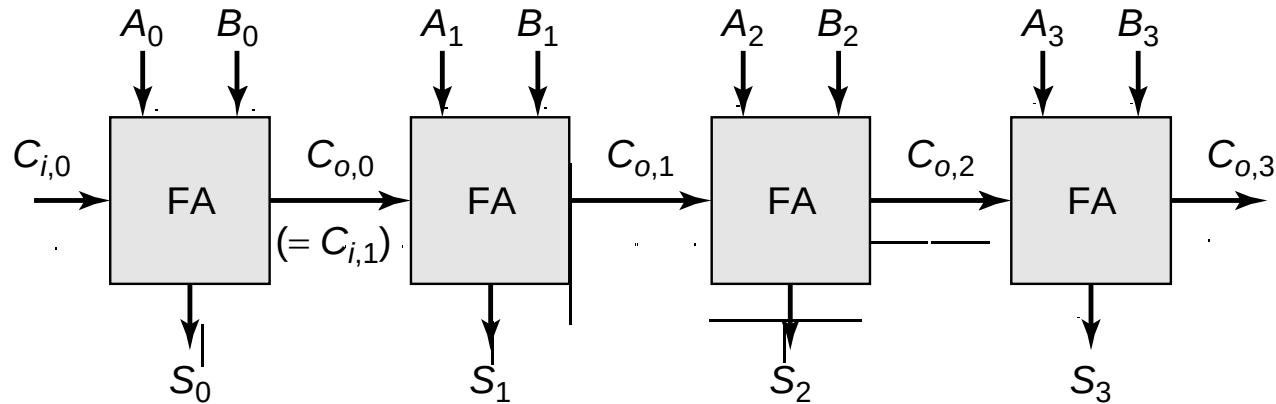
The Mirror Adder

- The NMOS and PMOS chains are **completely symmetrical**. A maximum of two series transistors can be observed in the carry-generation circuitry.
- When laying out the cell, the most critical issue is the minimization of the capacitance at node C_o . The reduction of the diffusion capacitances is particularly important.
- The capacitance at node C_o is composed of four diffusion capacitances, two internal gate capacitances, and six gate capacitances in the connecting adder cell .
- The transistors connected to C_i are placed closest to the output.
- Only the transistors in the carry stage have to be optimized for optimal speed. All transistors in the sum stage can be minimal size.

Transmission Gate Full Adder



The Ripple-Carry Adder



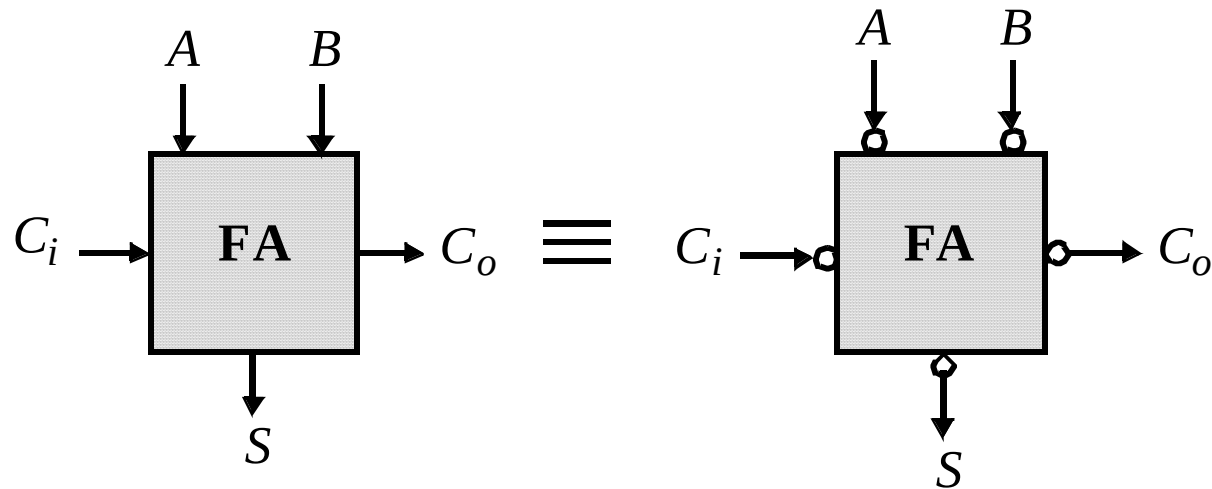
Worst case delay linear with the number of bits

$$t_d = O(N)$$

$$t_{adder} = (N-1)t_{carry} + t_{sum}$$

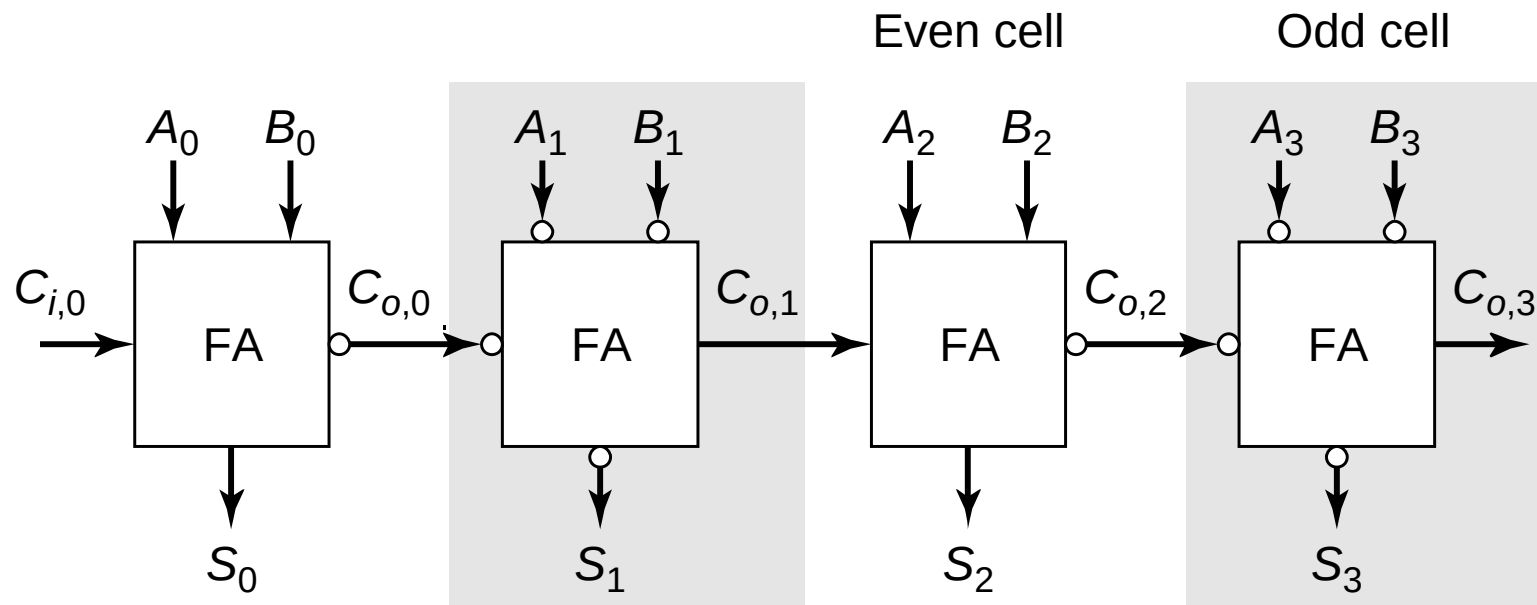
Goal: Make the fastest possible carry path circuit

Inversion Property



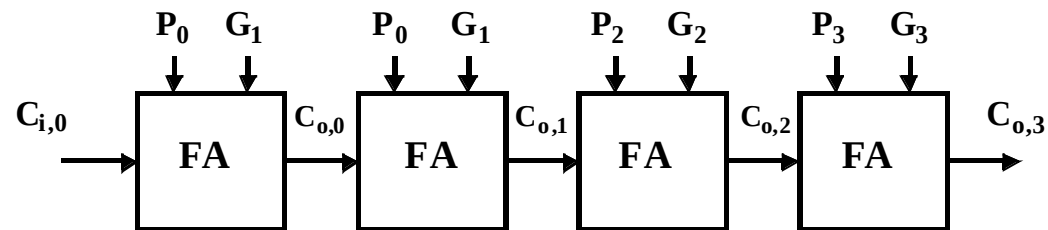
$$\bar{S}(A, B, C_i) = S(\bar{A}, \bar{B}, \bar{C}_i)$$
$$\bar{C}_o(A, B, C_i) = C_o(\bar{A}, \bar{B}, \bar{C}_i)$$

Minimize Critical Path by Reducing Inverting Stages

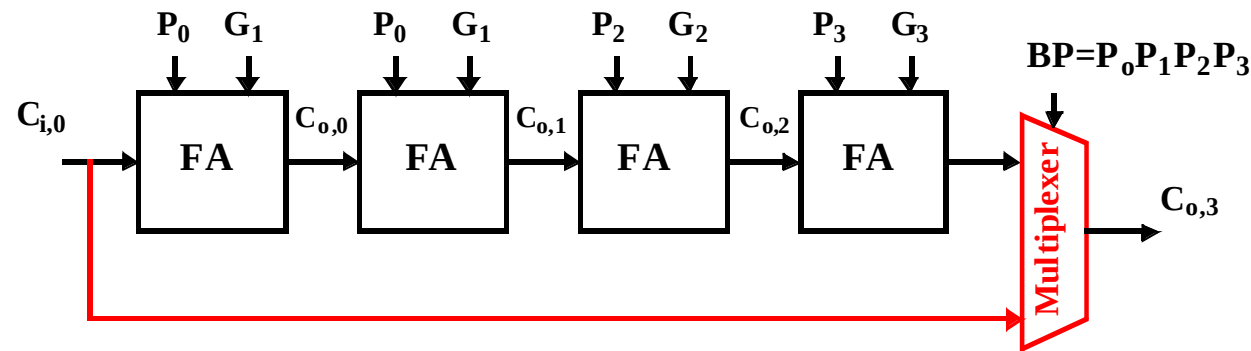


Exploit Inversion Property

Carry-Bypass Adder

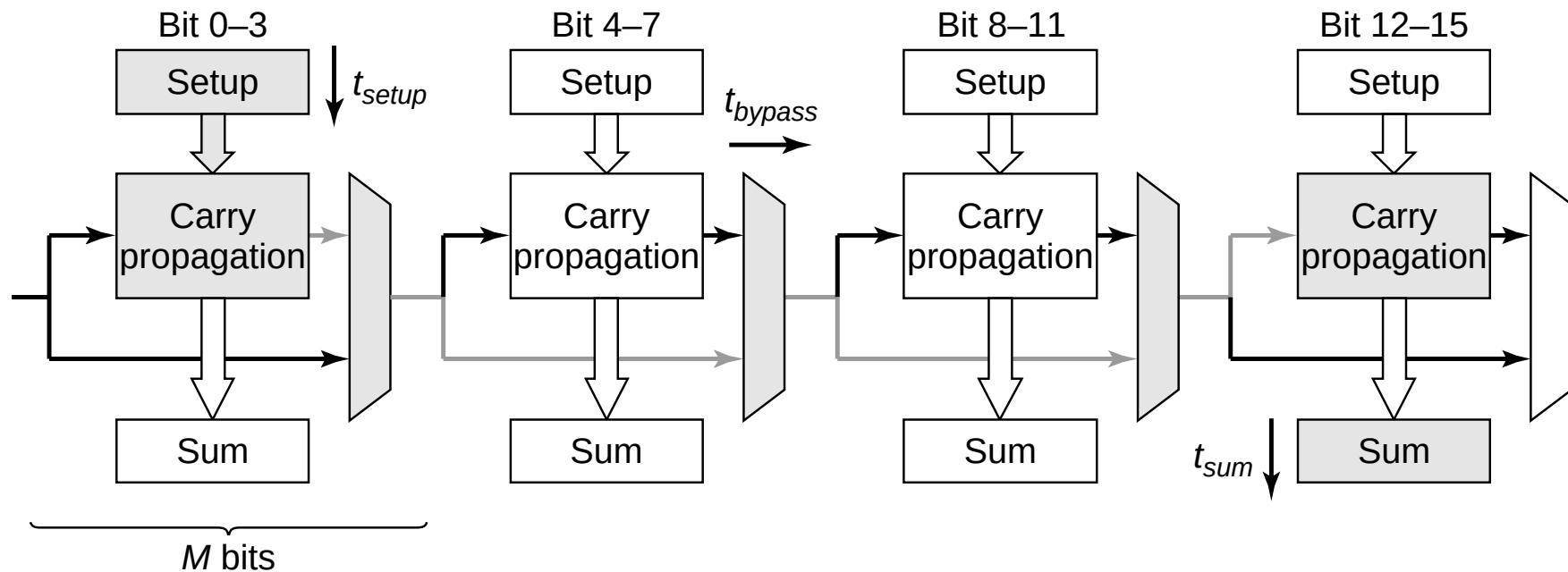


Also called
Carry-Skip



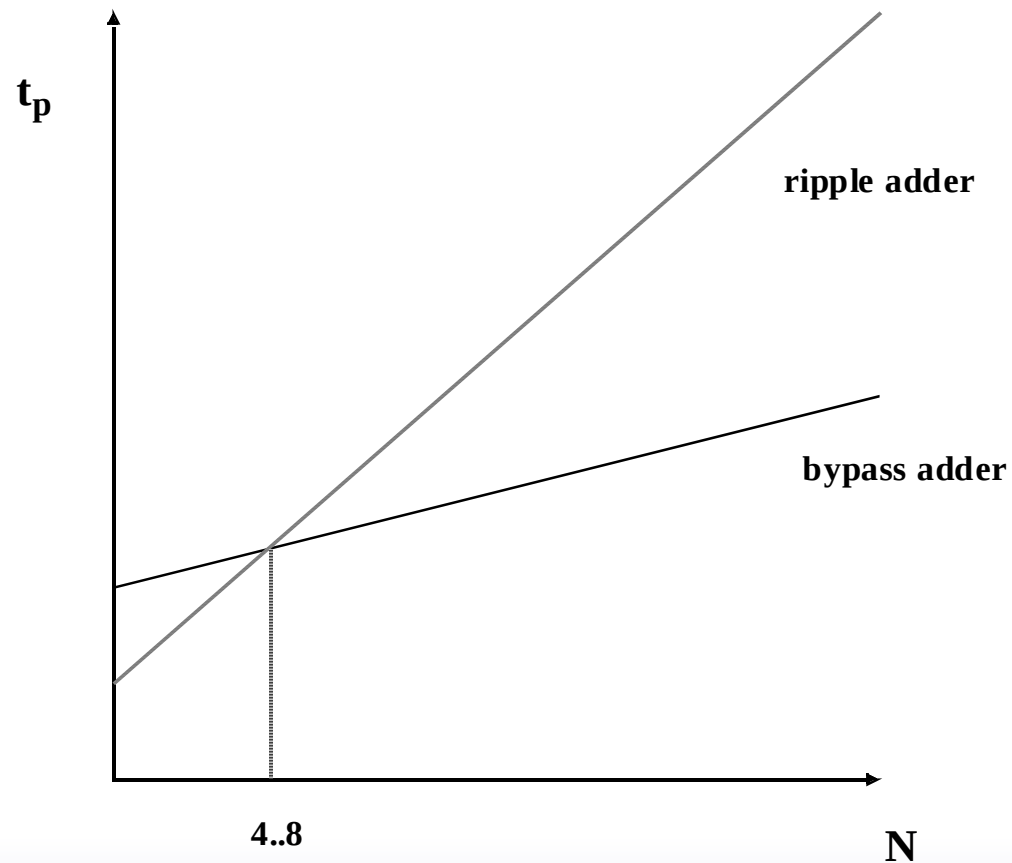
Idea: If (P_0 and P_1 and P_2 and $P_3 = 1$)
then $C_{o3} = C_0$, else “kill” or “generate”.

Carry-Bypass Adder (cont.)

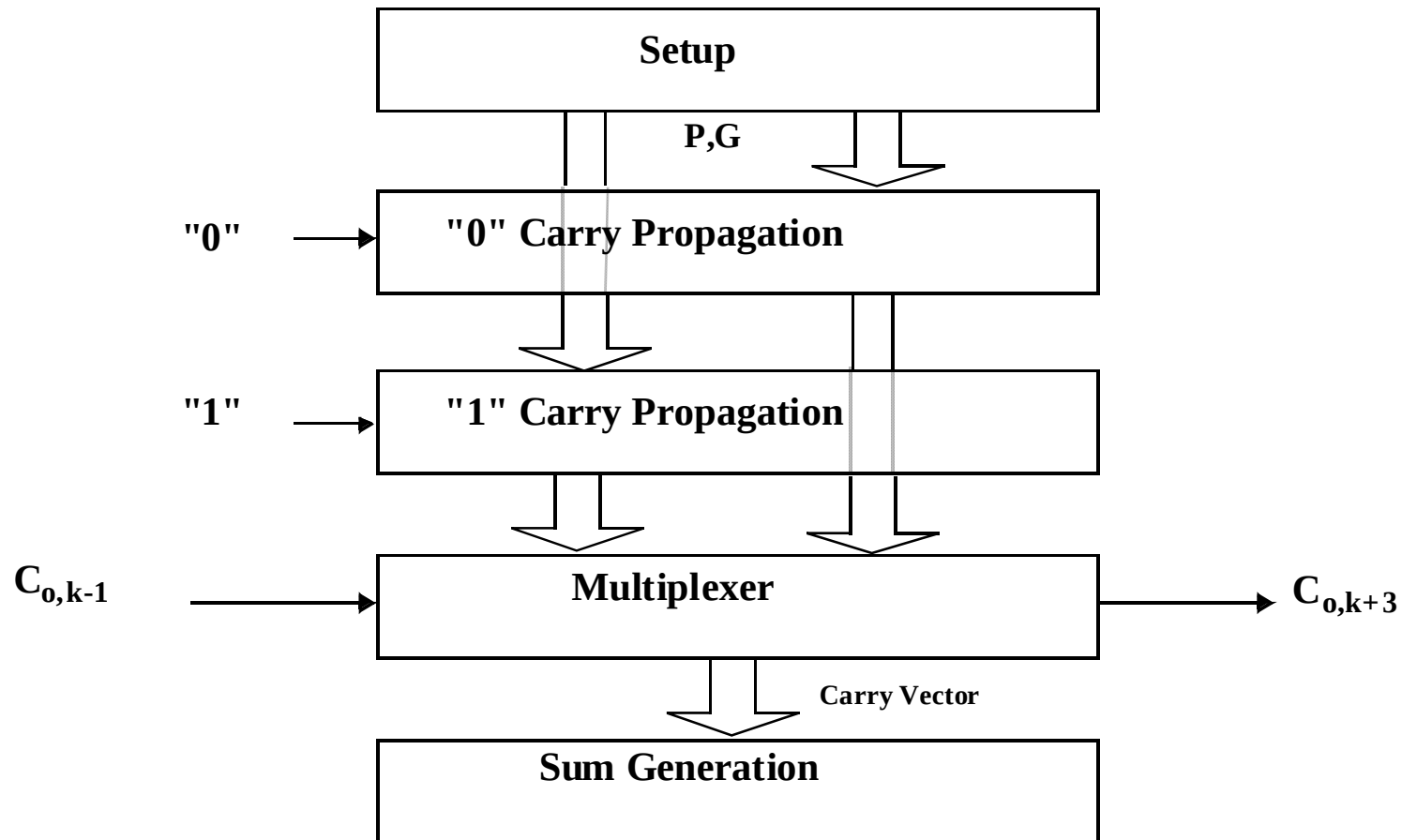


$$t_{adder} = t_{setup} + Mt_{carry} + (N/M - 1)t_{bypass} + (M - 1)t_{carry} + t_{sum}$$

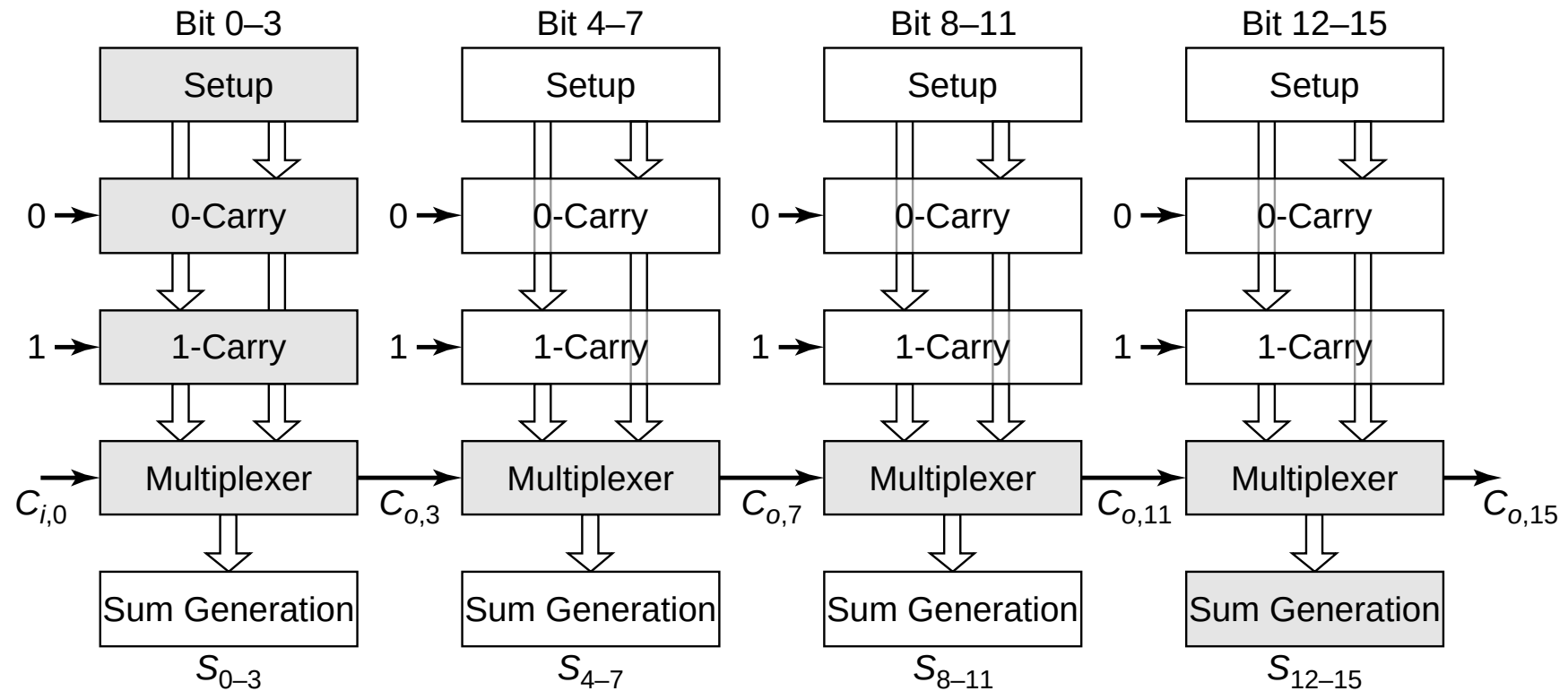
Carry Ripple versus Carry Bypass



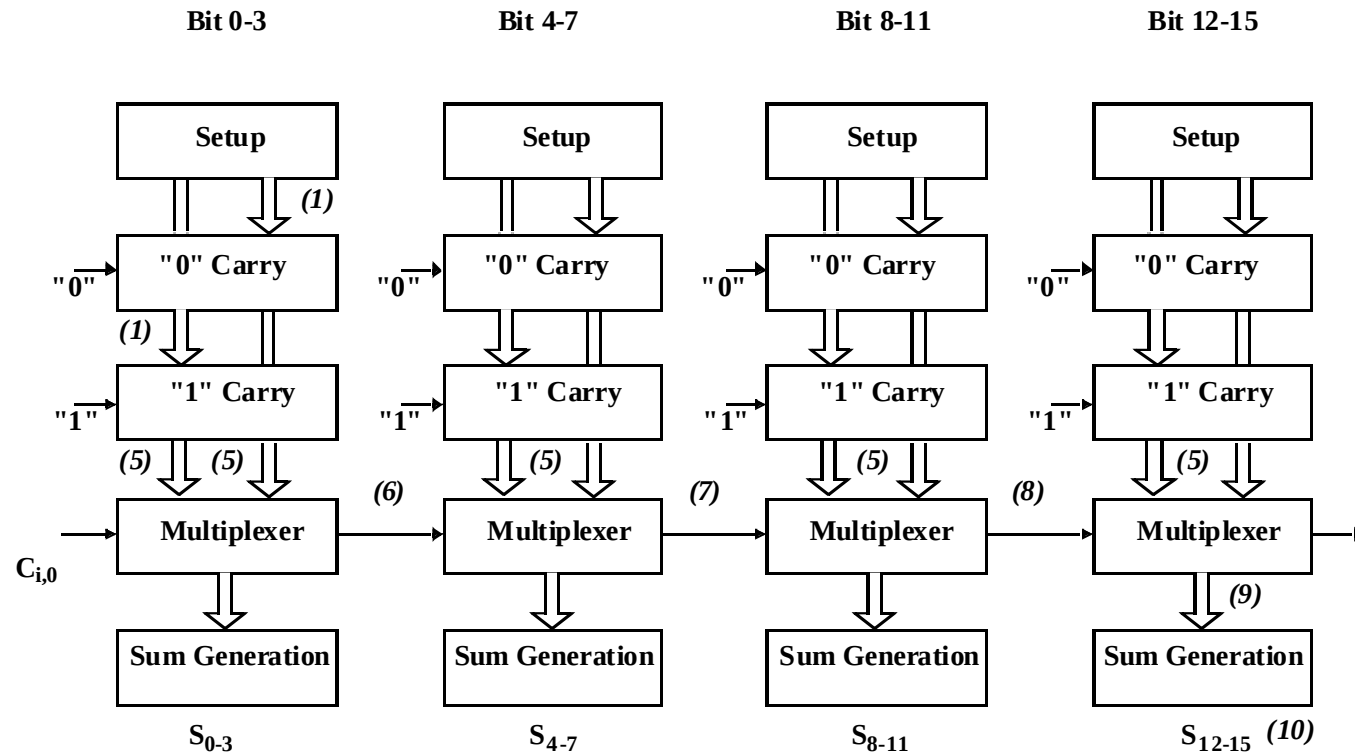
Carry-Select Adder



Carry Select Adder: Critical Path

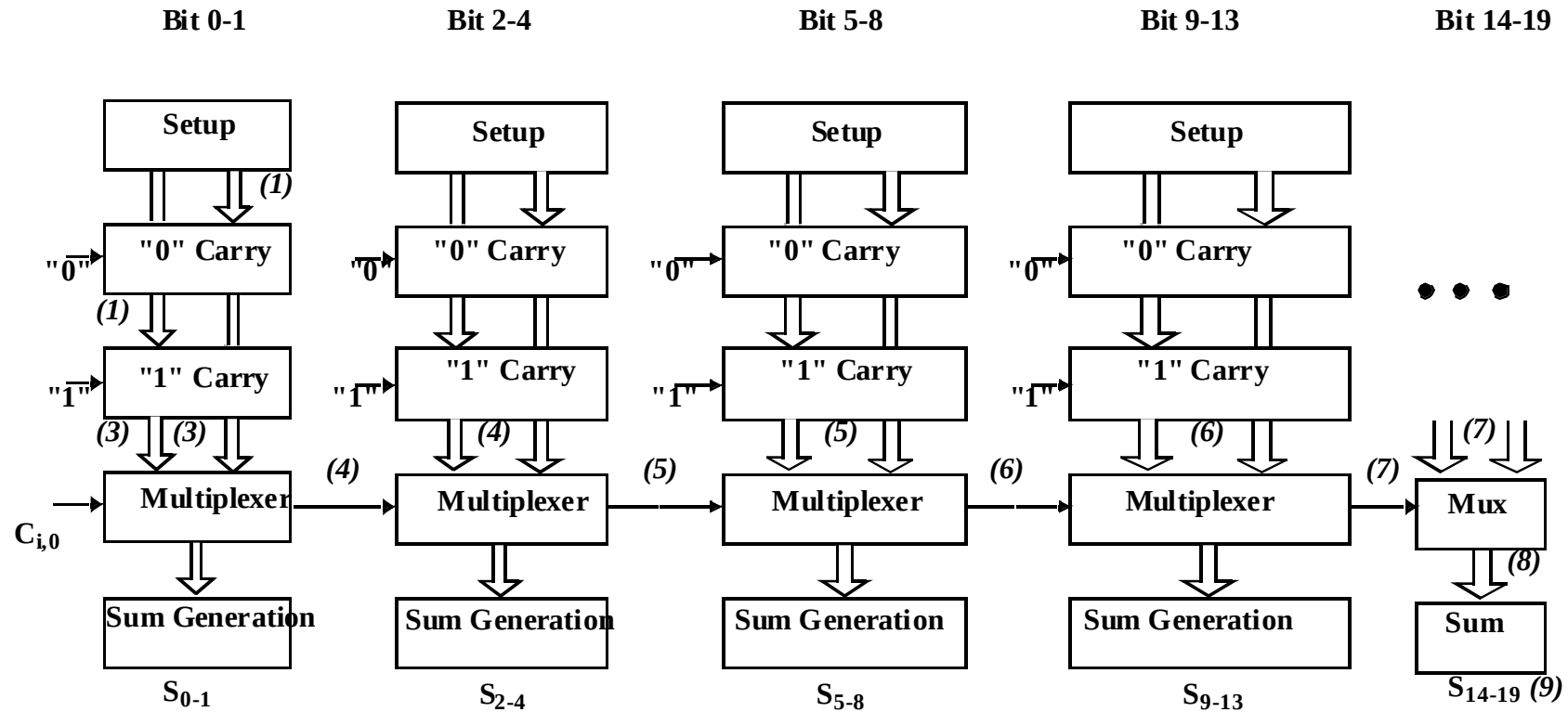


Linear Carry Select



$$t_{add} = t_{setup} + \left| M \right| t_{carry} + \frac{N}{M} t_{mux} + t_{sum}$$

Square Root Carry Select



$$t_{add} = t_{setup} + P \cdot t_{carry} + (\sqrt{2N}) t_{mux} + t_{sum}$$

Adder Delays - Comparison

